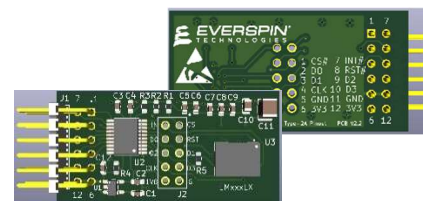


64 Mb MRAM PMOD

PERSYST STT-MRAM Persistent Memory

3.3V compatible, xSPI interface



Device Description

The EM064LX is a **PERSYST** family 64 Mb xSPI serial persistent memory. With an operating temperature range from -40°C to $+85^{\circ}\text{C}$, this industrial-grade device is suitable for both standard commercial settings and demanding industrial applications. The device itself supports Single, Dual, Quad and Octal protocols; however, this evaluation board specifically implements the Single, Dual, and Quad xSPI interfaces via a PMOD form factor compliant with the Digilent Type 2A interface. To ensure seamless integration, the board includes an onboard LDO and level shifters, adapting the EM064LX MRAM's native 1.8V VDD to comply with the standard 3.3V PMOD interface. Users now have the high performance and reliability of the production EM064LX in a convenient plug-and-play PMOD format.

Features

Industrial SPI MRAM PMOD

- Expanded SPI bus interface supporting
 - Single, Dual, and Quad xSPI protocol
- Virtually unlimited read and write operations, with no wear-out mechanisms or erase cycle requirements
- Data retention: 10 years minimum across temperature range
- JEDEC compliant: JESD251, JESD251-1 for QSPI
- Byte-addressable read and write access with no erase required
- Data integrity: No external ECC required.
- Low Power Modes (typical):
 - Standby $< 350\mu\text{A}$ Typical (64Mb)
 - Deep power down $\sim 50\mu\text{A}$ w/ exit time $< 100\mu\text{s}$
- SPI compatibility: NVSRAM, FRAM, NOR, Toggle MRAM
- Volatile and nonvolatile configuration settings
- 3.3V LVCMOS logic on all PMOD signals (internally level-shifted to the 1.8V MRAM core)
- Boot mode configurations
 - Boot in SPI or QSPI
- Software reset and hardware reset pin available
- 3-byte and 4-byte address modes
- Sequential (burst) read and writes
- Electronic signature
 - JEDEC-standard 3-byte signature
- Digilent PMOD Type-2A compliant SPI pinout with dedicated INT# and RST#
- RoHS compliant package:
 - 12-pin connector, 20 x 40 mm PCB
- Operating temperature range
 - From -40°C to $+85^{\circ}\text{C}$

NOR Compatible SPI, Quad xSPI Commands for Program/Erase and XIP, Security

- Execute-in-Place (XIP)
- Dedicated 256-byte OTP area outside main memory
 - Readable and user-lockable
 - Permanent lock with WRITE OTP command
- Erase is not required but the command is available
 - Chip / bulk erase and sector erase
 - Subsector erase 4KB, 32KB granularity
- Security and write protection
 - 16 configurable hardware write protected regions plus top/bottom select
 - CRC command to detect accidental changes to user data

Pin Functions

Signal Name	Type	Signal Level	Description
CS!	Input	3V3	Chip Select. Bus transactions are initiated with a HIGH to LOW transition. Bus transactions are terminated with a LOW to HIGH transition.
SCK	Input	3V3	Clock. Command, Address and Data information is transferred from controller to memory with respect to the rising or falling edge of the SCK. The clock is not required to be free running.
SD[3:0]	Input / Output	3V3	Serial Data Input/Output. Command, Address, and Data information is transferred on these signals during Read and Write transactions.
3V3	Power Supply	3V3	Power.
GND	Power Supply		Ground.
RST!	Input	3V3	Hardware Reset. When LOW, the memory will self-initialize and return the device to the ready state. DS and IO[7:0] are placed into the High-Z state when RST! is LOW. The RST! signal should not be allowed to float so a pull-up resistor is placed on the PMOD.
INT!	Output (Open Drain)	3V3	Interrupt. When LOW, the memory indicates that an internal event has occurred. This signal is intended to be used as a system level interrupt that indicates that an on-chip event has occurred. INT# is an open-drain output.

Table 2

Other Useful Information

For full SPI functionality, commands, timing diagrams and specifications please refer to the full EM064LX datasheet on the Everspin website. For clarity, the Pin Function table above references the ! symbol for the active low feature. In the EM064LX datasheet, the # is used for active low. This difference in nomenclature is intentional to distinguish between the native MRAM 1.8V VDD and the PMOD 3.3V levels.

Revision History

v1.1: April 2026 Revised version